The South African Economy

Chapter 1 — Why Do We Care About Economic Growth?

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This chapter is a standalone excerpt from the book *The South African Economy.* Cross-references to other chapters point to sections of the full manuscript.

Chapter 1

Why Do We Care About Economic Growth?

We need to understand what economic growth is before we ask whether growth is a good goal. In short, economic growth can be compared to any type of growth. Intellectual growth makes us more knowledgeable and more skilled, but cancerous growth can kill. We want the good type. But first, it is important to note that economic growth is an outcome, a thing we measure. It is the result of many moving pieces. We want to study how economic growth came to be and who the beneficiaries are. Do the benefits of growth accrue to the creators of growth? Should other people benefit from growth that they did not create? Should an agent, like a government, take from those that create growth and determine who gets what is taken? How do we get more growth? Are firms evil by manipulating markets and people to extract more resources? Or are they the engines of growth? We will try to answer some of these questions in the following chapters.

We measure the size of the economy by the goods and services created and sold. Economic growth measures how much those goods and services change over time. The more that is produced and sold, the higher economic growth will be.

Economists and statisticians measure the size of an economy using data on expenditures, incomes and production. These measurements are taken from markets; the market for traded goods such as food and cellphones, and factor markets such as labor and capital. Most of economics focuses on traded quantities and prices that clear (i.e., demand equaling supply). We want to measure how many new goods and services get created, at what price they are sold, and how expensive they were to produce. In economics, everything that is created is either sold or stored, and we have surveys to measure quantities and prices.

Because standardized global accounting practices exist, we can compare how much is produced locally versus abroad and estimate how efficient each country is at producing goods and services. We want to be efficient producers so that our output is competitive, i.e., producing better products at lower prices. The more our goods and services are demanded, the more inputs we require to produce them: land, workers, machines and other materials.

However, we can also try to include things that are not well captured by the market. This includes the value of our resources (services that the land provides such as bees in pollinating crops, or the opportunities that education provide in terms of coming up with new and inventive ideas to produce things better).

What is clear is that more demand for goods and services leads to a higher demand for the inputs needed to produce them. As a first and rudimentary measure, we care about economic activity because it is directly related to a higher demand for land, labor and machines.

Very simple formulas connect the different measurements of economic activity. We measure goods and services sold as follows:

$$Y_t = C_t + G_t + I_t + X_t - M_t + II_t + Stat_t \tag{1.1}$$

where Y is gross domestic product (or GDP), C is household consumption (typically made up of durables, non-durables and semi-durables), G government consumption (includes the wage bill of government and the purchases made by the government), I is total investment by both the private sector and the government, X is exports of goods and non-factor services, M is the imports of goods and non-factor services, II is the change in inventories and $Stat$ is a statistical discrepancy, or measurement error when relating GDP accounts. t is an index that accounts for time (in this case years).

In national accounts, we can write GDP in terms of nominal values, constant price or chain-linked values. Nominal values reflect the value of goods and services. This is just a quantity multiplied by a price. Constant price GDP reflects the value of goods and services in a specific year. Thus any change in constant price GDP reflect changes in quantities and not prices.

To consume goods we need them produced (either locally or from abroad). We use goods from different sectors. Total GDP from the sectoral side is then the sum of agriculture (AGR), industry (IND) and services (SRV) plus net indirect taxes and subsidies (NIT):

$$Y_t = AGR_t + IND_t + SRV_t + NIT_t \tag{1.2}$$

Finally, the value of goods and services reflect the input costs and the profits. We can think of the value of a cellphone reflecting labor costs, or the wage bill, which is simply the number of people employed (N) multiplied by the wage that they earn (W), but also reflecting the capital costs, which is the rental rate of capital (R) multiplied by the capital used, e.g., machines or buildings (K):

$$Y_t = W_t N_t + R_t K_t + NIT_t \tag{1.3}$$

We can close the accounts by noting that value-added for each industry is equal to input costs. For agriculture, as an example we can write this as:

$$AGR_t = W_t^{AGR} N_t^{AGR} + R_t^{AGR} K_t^{AGR} \tag{1.4}$$

Try not to get too distracted by the formulas. They are meant to keep book of the things that are produced and sold. Later in the book we will build out theories for how things are produced and the policies that affect consumption and production.

1.1 How Do We Compare in Terms of Producing Goods

Now we have a standard measure for the value of goods produced, which means we can compare South Africa to its peers. The comparison roughly tells us how "well off" the average South African is to countries who are more developed and countries who we think have similar levels of development. It does not tell us whether a South African is more content with what she has compared to someone in the United States of America. But it does tell us a bit about how productive we are and how good we are about the allocation of resources.

Let's start by asking how big the "pie" is that we can divide between ourselves. We do this by dividing GDP by the number of people in each country. GDP per capita tells us how much output is produced per person. Note that this is roughly an average, since some people produce more goods and services than others. The larger the "pie" (in our case the economy), the more there is to divide!

Figure 1.1 shows that South Africa (the thick line) has stagnated since 2007. GDP produced per person in 2023 is roughly the same as it was in 2007. Real GDP growth and population growth have been roughly equal, implying that either capital accumulation has contributed very little or that the efficiency with which South Africa produces is falling. Both turn out to be true. Later we will look at the role of electricity constraints, which has been a major factor in constraining economic activity.

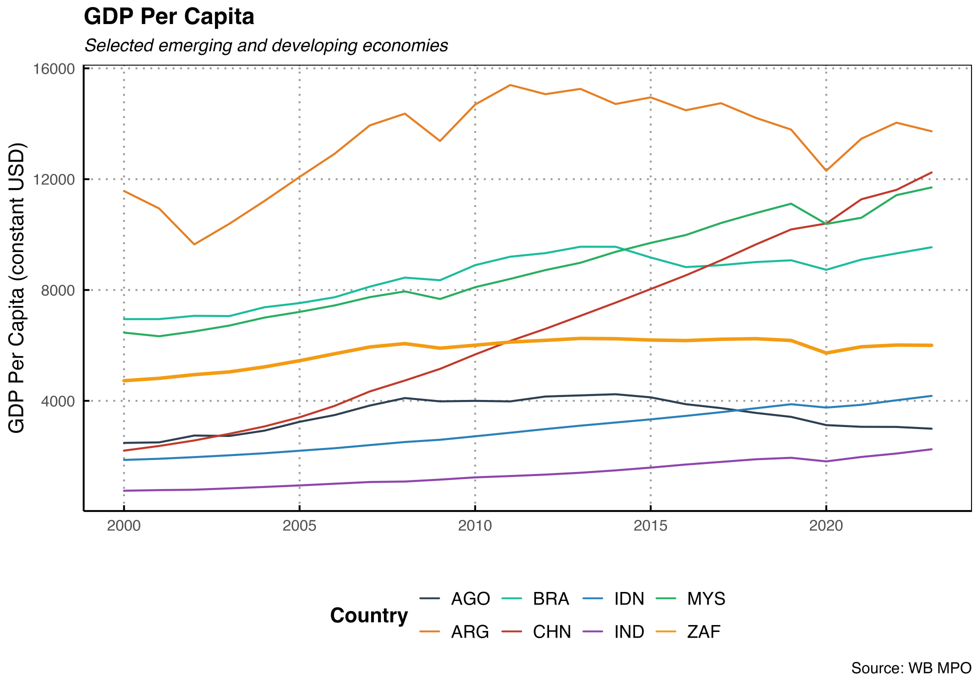

Figure 1.1: GDP Per Capita

It is also clear that South Africa's share of global output is small and shrinking. In Figure 1.2 we plot South African GDP (in real constant dollars) as a share of world GDP.

It is striking, although not surprising, that high-income countries make up the bulk of total GDP. India and China are the major exceptions, and most of the shift in shares over time has been driven by them. Both countries are growing rapidly in GDP per capita terms, and consequently their share in global output is rising. But what drives their growth? And can it last? These are important questions, because superficial observers often point to China and India as models to emulate without understanding what actually makes them tick, or what makes them vulnerable.

What Really Drives Growth in China and India?

China and India together account for over a third of the world's population and an ever-growing share of global output. It is tempting to point at them and say: "See, state-led development works." But the reality is far more nuanced.
China's growth miracle began not with more state control, but with *less*. Deng Xiaoping's reforms after 1978 unleashed market forces: special economic zones, private enterprise, price liberalization, and openness to foreign investment. The Communist Party did not plan China's growth. It got out of the way in the areas that mattered

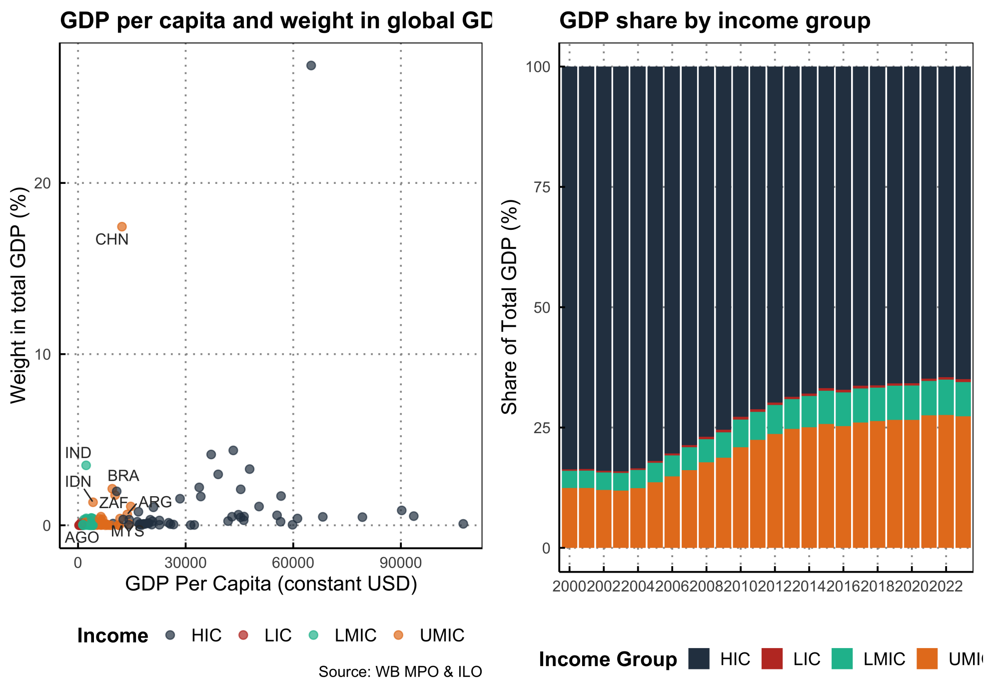

Figure 1.2: Weights of total GDP

most. Chinese growth has been driven by extraordinarily high savings rates (above 40% of GDP), massive capital investment, an educated and disciplined labor force, and rapid technology adoption from abroad. Culturally, Confucian values (hard work, education, thrift, family obligation, and respect for order) created a population primed for economic discipline. China's people drove the miracle; the state merely removed some of the barriers and directed infrastructure.

But China's model is now showing cracks. Decades of over-investment have produced ghost cities, a property crisis, and a debt-to-GDP ratio exceeding 300%. The one-child policy created an aging population that will shrink the labor force for decades. Total factor productivity growth has slowed sharply since 2010. And the very authoritarianism that enabled fast infrastructure decisions also suppresses the innovation, free inquiry, and creative destruction that sustain long-run growth. China may be approaching a middle-income trap of its own making.

India's growth story is different. India liberalized in 1991, dismantling the "License Raj" that had strangled private enterprise for decades. Growth since then has been driven by services (IT, finance, telecommunications) and a massive demographic dividend (a young, growing labor force). India's democratic institutions, English-language advantage, and entrepreneurial culture have powered its rise. But India's growth is deeply uneven: infrastructure remains poor, the informal sector employs over 80% of workers, and bureaucratic red tape still chokes business formation. India grows *despite* its government, not because of it.

What do both countries share? Three things that South Africa lacks. First, **cultural cohesion**: China is 92% Han Chinese; India, for all its diversity, has deep civilizational continuity and shared religious traditions that bind communities together. Both societies have strong family structures that substitute for weak state institutions. Second, **a work ethic rooted in culture**: in both countries, hard work, savings, and education are deeply valued, not because the state mandates it, but because families and communities demand it. Third, **high savings rates**: both countries save and invest far more than they consume, providing the capital for growth without dependence on foreign borrowing. The lesson for South Africa is *not* that we need a stronger state. China's state did not create growth; it released it by stepping back. India grows in spite of its bureaucracy. The lesson is that growth requires a disciplined, cohesive, hardworking population that saves and invests and institutions that let them do so freely. These are ultimately questions of culture, values, and character. They are questions of the soul.

1.2 Growing an Economy Is Good for Generating Jobs

Chapter Hypothesis

Economic growth driven by productivity improvements and capital accumulation—not government spending—creates sustainable employment. Countries that grow faster reduce unemployment; countries that stagnate, like South Africa, trap millions in joblessness.

This is a very simple hypothesis that we will unpack in this book. However, as we will see, the relationship between economic growth and jobs is more nuanced than expected. Is growth related to employment growth? If you say no, you fall into the camp that claims there is such a thing as jobless growth. If you say yes, you are making the association that growth and jobs are correlated.

Those who say no, i.e., that growth can occur without creating jobs, will typically argue from a perspective that favors larger government intervention to create employment directly. Those who say yes will argue that growth itself is the mechanism through which jobs are created, and that we should focus on removing barriers to growth.

It is tempting to start the analysis with the elasticity of growth to employment. To make this point clear, we plot GDP per capita against unemployment rates across countries for the year 2023. Visually, higher-income countries tend to have much lower unemployment rates. Is it just capitalists that benefit from growth? It does not seem so, growth appears to benefit everyone.

In Figure 1.4 we plot the correlation between employment growth and GDP growth. This is not causation. We are not proving that growth leads to employment, only showing the relationship between the two. For the majority of countries, the correlation between economic growth and employment growth is positive. There are a few countries (small relative to the total sample) where the relationship is negative. What causes growth? Part of it is employment growth. What causes employment? Part of it is economic growth. We will disentangle these effects in Chapter 3.

It is, however, clear that the relationship between employment growth and GDP growth in South Africa is particularly strong. Positive growth rates are associated with positive employment growth, which means that when South Africa's growth stalls, jobs disappear.

Unfortunately, these correlations capture only short-run effects. In the long run, employment growth is not really a function of economic growth. We can demonstrate this by looking at the demand and supply of labor through the unemployment identity:

$$N_t = (1 - UNR_t) \cdot PR_t \cdot WPOP_t \tag{1.5}$$

where N is the number of people employed, UNR is the unemployment rate, PR is the

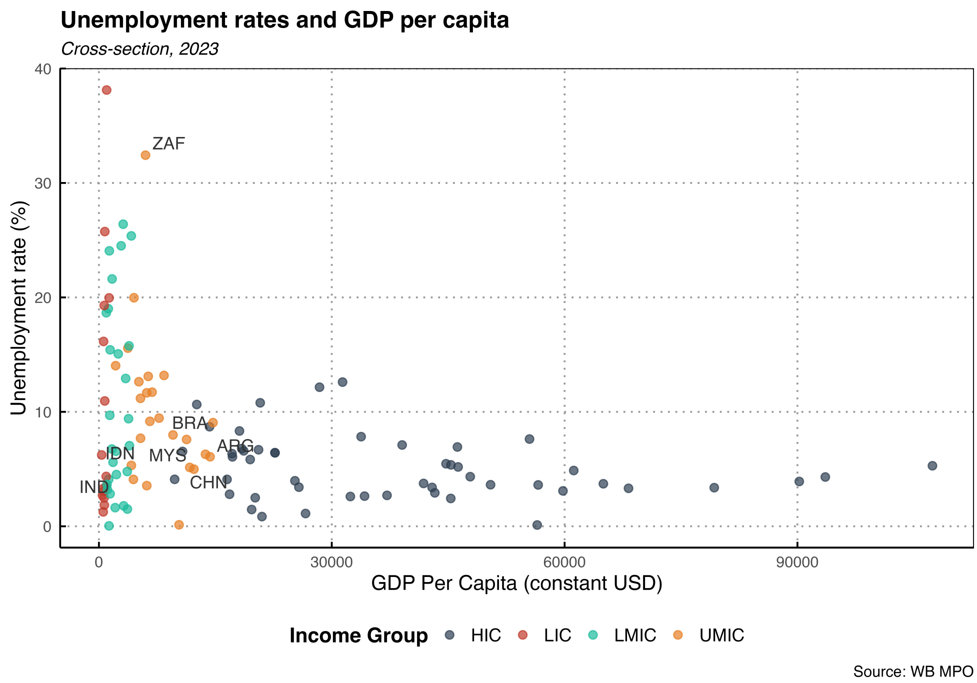

Figure 1.3: Unemployment rates and GDP per capita

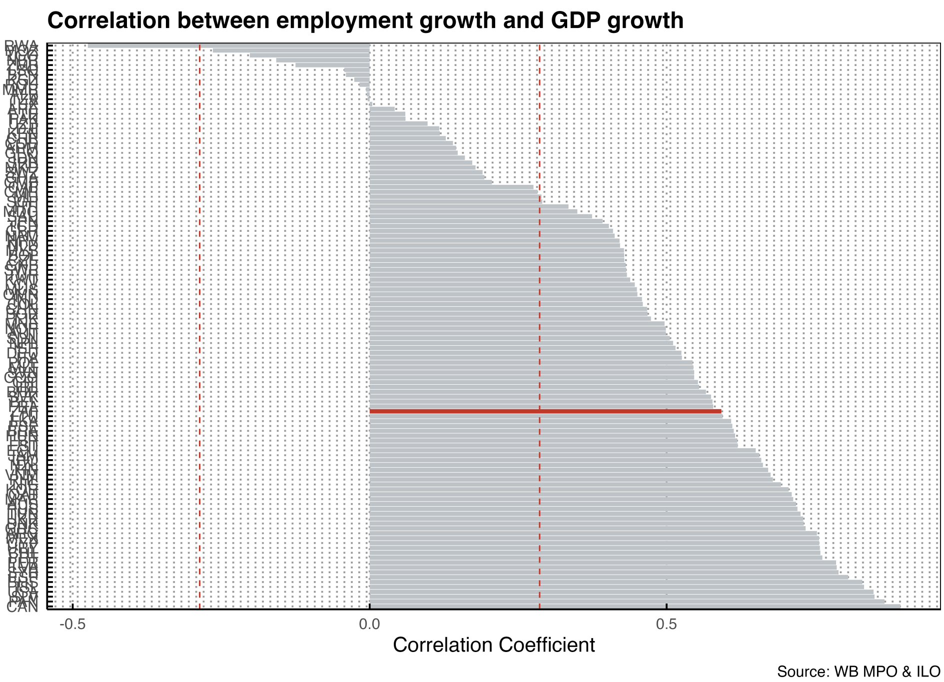

Figure 1.4: Correlation of employment growth to GDP growth

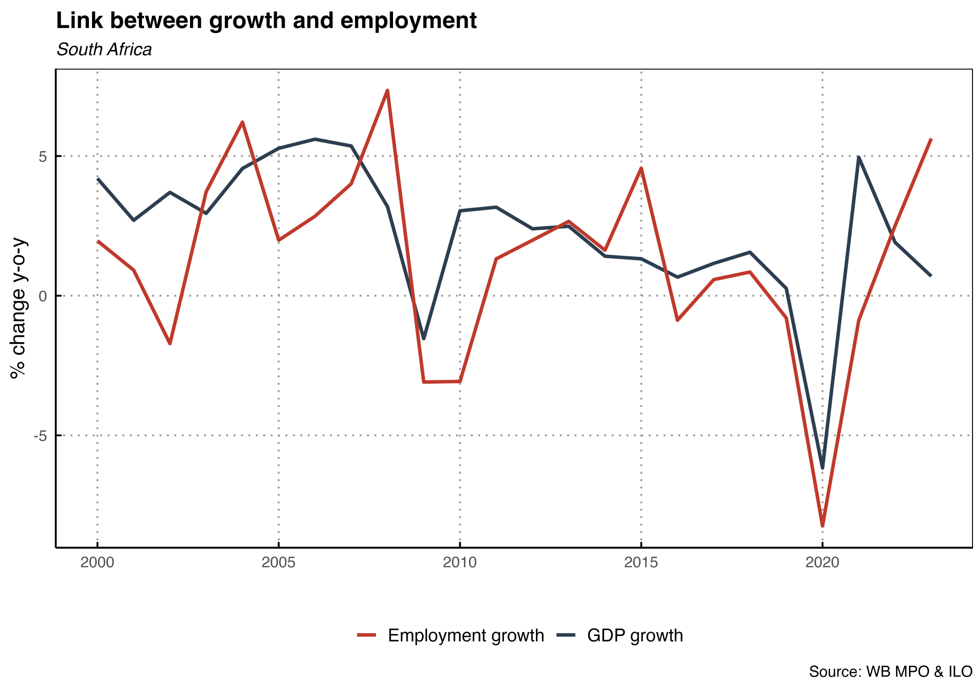

Figure 1.5: South Africa's employment and GDP growth

labor force participation rate and $WPOP$ is the working age population.

Assume that the unemployment rate is equal to zero (more realistically it fluctuates around a fixed point), that the participation rate fluctuates around a fixed point and that the working age population growth rate is in line with population growth. GDP growth does not enter this relationship. It is pointless to talk about jobless growth since by design, long run employment growth is mainly a function of population growth.

In the the labor market chapter we will discuss how jobs get created in great detail. A short summary: A pie can only grow if people are able to consume it. I.e., the diffusion of growth has to trickle down to everyone, even if not in equal measure.

In many countries, particularly South Africa, the unemployment rate is very high and not too many working age people participate in the labor market. The reason why unemployment rates are high might be the same reason why GDP growth is low. We call these reasons distortions. Something, e.g., a bad policy, is causing unemployment to be high. Even in many low income countries unemployment is low because people cannot afford to be unemployed (many work in the informal market). The labor market only has a few margins of adjustment when we see policies or obseve shocks: (i) Either unemployment increases when wages do not adjust, when firms cannot change output and when workers do not move to informal jobs, (ii) or when wages do not adjust (think union bargaining, minumum wages), then informality might increase keeping unemployment constant; (iii) or wages adjust so that informality and unemployment remain the same, or (iv) firms get destroyed, destroying both job creation and wage remuneration, resulting in high levels of informality.

Removing the distortion will reduce the unemployment rate. Note that the unemployment rate is just the difference between the demand for labor (N) and the supply of labor ($PR \cdot WPOP$). In the limit, we can never grow by more than the supply of labor, and in the limit, the supply of labor is constrained by the working age population.

So, there is a relationship between growth and employment, but this is mainly during business cycle movements. In the long run, there are factors that lower growth and create unemployment, and it is our job to identify those distortions and wedges.

However, economic growth is also related to better jobs. Better jobs bring people out of poverty and lead to better welfare. To say that better jobs are not associated with economic growth is false, unless all growth comes from automation, something that has not yet happened in our lifetime, or that of our forebears. In the next chapter we will demonstrate that a more productive workforce (a more educated workforce) contribute significantly to developed country growth. A more educated workforce also receives higher incomes than those with lower levels of education.

So here is the bottom line. If we care about people's wellbeing then we should care about economic growth. Not growth for growth's sake, but growth because it is the mechanism through which more and better jobs are created, incomes rise, and poverty falls. The data shows this clearly: higher GDP per capita is associated with lower unemployment, and employment growth and GDP growth move together. The question, then, is not *whether* to pursue growth, but *how*. And the answer to that question depends enormously on which economic framework you believe in, because different frameworks lead to radically different policies, and those policies shape whether a society flourishes or stagnates. This is what we turn to next.

1.3 On Different Economic Schools of Thought and What Ultimately Drives Growth

Economics is not a neutral science practiced in a vacuum. The school of thought that a government subscribes to, consciously or not, determines the policies it pursues, and those policies determine whether an economy grows or shrinks, whether jobs are created or destroyed. If a government leans Marxist, as South Africa's ANC alliance seems to through its ties to the South African Communist Party and COSATU, then we should expect policies that emphasize redistribution over production, state ownership over private enterprise, and regulation over market freedom. If a government favors smaller government and market competition, as in Singapore or Botswana, we should expect lighter regulation, fiscal discipline, and policies that let the private sector drive employment.

But schools of thought do not only operate at the level of government. They shape how *individuals* think and act. A person raised in a Marxist intellectual tradition sees the world through the lens of exploitation and class conflict. Work is something extracted from you, profit is theft, and the system is rigged. That worldview discourages entrepreneurship, savings, and personal responsibility because the individual believes the structure, not the self, must change. A person raised in a tradition that values free exchange and personal agency sees opportunity where the Marxist sees oppression; work is how you build, savings are how you invest in the future, and your outcomes are largely a function of your choices. These are not just policy differences. They are differences in how a person relates to effort, to risk, to other people, and ultimately to the question of what makes a good life. The school of thought that dominates a society's universities, media, and political discourse shapes not just the laws on the books but the character of its citizens.

This is why the debate matters so profoundly. It is the most consequential question a country faces, not only because it determines tax policy, labor regulation, trade policy, the size of the public sector, and the role of state-owned enterprises, but because it shapes the beliefs and behaviors of millions of individuals whose daily choices *are* the economy. Below I outline the three major schools of thought, their view of growth, and the *policies* that follow from each. The reader can then judge which set of policies most resembles what South Africa has been doing, and whether the outcomes match the promises.

1.3.1 Marxist Perspective on Economic Growth

Marxist economics sees economic growth as inherently linked to capital accumulation and the contradictions within capitalism.

Key Determinants:

- Capital Accumulation: Growth occurs as capitalists reinvest surplus value extracted from labor into expanding production.
- Labor Exploitation: The key driver of profits (surplus value) is the exploitation of workers where capitalists pay workers less than the value they produce.
- Technology & Organic Composition of Capital: Increasing capital intensity (machinery replacing labor) can lead to falling profit rates over time.
- Crisis Tendencies: Overproduction, underconsumption, and profit rate declines create cyclical crises that periodically disrupt growth.
- Class Struggle & Institutional Change: Workers' resistance and changes in labor-capital relations (e.g., unions, state intervention) shape long-term growth trajectories.

Growth Dynamics:

- Growth is fundamentally unstable and prone to crises.
- The economy is driven by profit-seeking behavior, but contradictions (falling profit rates, class struggle) limit long-term sustainable growth.
- Technological progress increases productivity but does not always ensure stable or equitable growth.

Typical Policies:

- **Goal: Equity and redistribution:** Nationalization of key industries (mining, banking, energy), land expropriation, wealth taxes, and state-directed investment.
- **Goal: Labor empowerment:** Strong union protections, centralized wage bargaining, restrictions on firing, and minimum wage floors.
- **Goal: State control of the economy:** Large state-owned enterprises, industrial policy driven by government rather than markets, trade protectionism to shield domestic producers.

The appeal is obvious: if workers are exploited, protect them; if capital is concentrated, seize it. But the track record is devastating. Every country that has pursued these policies aggressively such as the Soviet Union, Venezuela, Zimbabwe, has destroyed wealth rather than redistributed it. South Africa's partial adoption of this framework through BEE, rigid labor laws, bloated state-owned enterprises (SOEs), and hostility to foreign capital has produced exactly the outcomes Marxist theory predicts for capitalism (stagnation, unemployment, inequality) while claiming to oppose them.

1.3.2 Neo-Classical Perspective on Economic Growth

Neo-Classical economics, especially through Solow's Growth Model, explains growth primarily through factor accumulation and technological progress.

Key Determinants:

- Capital Accumulation: More investment in capital stock leads to higher output. Not permanently though.
- Labor Growth: Population growth and labor force participation impact economic output.
- Technology (Total Factor Productivity, TFP): The main long-run driver of growth is exogenous (or its endogenous variants) technological change.
- Diminishing Returns & Convergence: Countries with lower capital per worker should grow faster and catch up to richer economies (conditional convergence).

Growth Dynamics:

- Growth follows a steady-state path determined by savings, depreciation, and technological progress.
- Markets are assumed to be self-correcting, with growth being supply-driven (long-term growth depends on increasing productivity and efficiency).
- Government intervention is seen as secondary, with policy focused on improving efficiency and innovation.
- Endogenous Growth Theory (Romer, Lucas): Later extensions emphasize knowledge, human capital, and innovation as engines of growth, making technological progress endogenous.

Typical Policies:

- **Goal: Growth through efficiency:** Lower and simpler taxes, deregulation of product and labor markets, trade liberalization, secure property rights, and independent central banking.
- **Goal: Human capital:** Investment in education and skills (supply-side), competitive schooling, and portable training vouchers rather than centralized programs.
- **Goal: Innovation:** Patent protection, R&D tax incentives, university-industry linkages, and openness to foreign technology and investment.
- **Goal: Fiscal discipline:** Government spending as a fixed share of GDP, balanced budgets over the cycle, and debt sustainability.

The neo-classical framework asks government to do less, but to do it well: enforce contracts, protect property, provide public goods that markets undersupply (defence, basic infrastructure, rule of law), and then get out of the way. Countries that have followed this prescription (see for example South Korea, Chile, Botswana, the Baltic states) have achieved sustained growth and dramatic poverty reduction. The framework's weakness is that it can underestimate the role of demand in the short run and the political difficulty of structural reform. But its long-run predictions are the best supported by the data, as we will see in Chapter 2.

1.3.3 Post-Keynesian Perspective on Economic Growth

Post-Keynesian economics challenges Neo-Classical assumptions, emphasizing demand-driven growth, income distribution, and financial instability.

Key Determinants:

- Effective Demand & Investment: Growth depends on aggregate demand, particularly investment decisions by firms.
- Income Distribution: Wage-led vs. profit-led growth; whether higher wages stimulate demand or higher profits fuel investment affects growth dynamics.
- Role of Finance & Debt: Credit creation and financial markets influence long-run growth and stability (Minsky's financial instability hypothesis).
- Path Dependency & Hysteresis: Economic growth is not necessarily convergent; recessions and shocks can have long-term scarring effects.

Growth Dynamics:

- Growth is driven primarily by demand, meaning investment decisions, government spending, and trade dynamics matter more than just technology and factor accumulation.
- Instability is a feature, not a bug: financial cycles, demand fluctuations, and income inequality shape long-term growth.
- Unlike Neo-Classical models, there is no automatic convergence—countries can be trapped in low-growth regimes.

Typical Policies:

- **Goal: Demand stimulation:** Counter-cyclical fiscal policy (increase spending during recessions), public works programs, and active industrial policy to direct investment.
- **Goal: Wage-led growth:** Minimum wages, collective bargaining support, and income redistribution to boost consumption by lower-income households.
- **Goal: Financial stability:** Regulation of financial markets, capital controls, and central bank mandates that include employment targets alongside inflation.
- **Goal: Reducing inequality:** Progressive taxation, social transfers, and public provision of housing, healthcare, and education.

The post-Keynesian framework contains important insights such as demand does matter, financial instability is real, and recessions can leave permanent scars. But here is the trap: every policy prescription in this framework requires a *larger* government. More spending, more regulation, more redistribution, more state direction of investment. For a country like South Africa, which already has an oversized, inefficient, and corrupt public sector (as Chapters 4 and 5 will document), demand-led policies are not the solution, but are the problem dressed in different clothes. Expanding government spending when the government cannot deliver clean water to its own citizens is not stimulus. It is waste. Raising minimum wages when 33 per cent of the labor force is unemployed does not boost demand, it destroys jobs for the most vulnerable. The post-Keynesian prescription assumes a competent, honest state. South Africa does not have one. Applying demand-side medicine to a supply-side disease will make the patient worse.

The table above is not an academic exercise. It is a mirror. The reader should ask: which column looks most like South Africa? Which policies has the government pursued? And what have the outcomes been? In Chapter 2 we will explore the actual drivers of economic growth (e.g., capital, labor, productivity, intermediate inputs) and let the data speak. The data, as we will see, supports the neo-classical view of growth most convincingly. But it also reveals something deeper: that the efficiency with which a society produces depends not just

Table 1.1: Comparison of economic schools of thought on growth

Theory	**Key Growth Driver**	**Stability**	**Role of Government**	**Typical Policies**	**Primary Goal**
Marxist	Capital accumulation, class struggle	Crisis-prone	Revolution; state ownership	Nationalization, land reform, wealth taxes, union power	Equity, worker control
Neo-Classical	Capital, labor, technology (TFP)	Stable, convergent	Minimal; enforce rules, provide public goods	Low taxes, deregulation, trade openness, property rights, fiscal discipline	Growth, efficiency
Post-Keynesian	Effective demand, investment, finance	Unstable, path-dependent	Strong; demand management, redistribution	Public works, minimum wages, industrial policy, financial regulation	Demand, equality

on its policies but on the honesty, discipline, and character of its people.

The Savings Channel: Why Growth Begins with the Soul (Proposition 2)

The Solow model shows that a country's steady-state income depends critically on its savings rate. But what determines the savings rate? Standard economics points to interest rates and demographics. The trichotomous model developed in Chapter 9 adds a deeper driver: the soul state. Proposition 2 shows that the curvature of the utility function, $\sigma(S)$, rises with the soul state S. An agent with a high soul state is more easily satiated; she does not need to consume everything today. She saves more, invests more, and her society reaches a higher steady-state GDP per capita. South Africa's national savings rate hovers around 15% of GDP, compared to 30–45% in East Asia's high-growth economies. The gap is not just about tax incentives or pension policy. It reflects a society oriented toward immediate consumption rather than patient stewardship—a low-soul-state equilibrium (see Appendix .18 for the full derivation).

Economics begins with incentives. It ends with the soul.

1.4 Chapter Summary

- Economic growth is measured by the change in goods and services produced over time. GDP per capita is the relevant measure for living standards.
- South Africa has stagnated relative to its peers since 2008. Countries that were poorer than South Africa two decades ago have overtaken it.
- Higher-income countries have lower unemployment rates. Growth and employment are positively correlated across countries, confirming that growth is the primary engine for job creation.
- Three schools of thought compete to explain growth: Marxist (redistribution and state control), neo-classical (free markets and productivity), and post-Keynesian (demand management). South Africa's policies have increasingly leaned toward the first and third, with predictable results.

- The neo-classical framework, which emphasizes capital accumulation, human capital, TFP, and institutional quality, best fits the cross-country evidence. Chapter 2 develops this in detail.
- South Africa's low savings rate (around 15% of GDP) constrains capital accumulation. The trichotomous model connects this to the soul state: satiated agents save more, enabling higher steady-state income (Proposition 2).